

A Few Tips for Scaling Up Web Performance

Every millisecond counts when it comes to loading Web pages and their responsiveness. It has become critical to optimise the performance of Web applications/pages to retain existing visitors and bring in new customers. If you are eager to explore the world of Web optimisation, then this article is the right place to start.



The World Wide Web has evolved into the primary channel to access both information and services in the digital era. Though network speed has increased many times over, it is still very important to follow best practices when designing and developing Web pages to provide optimal user experiences. Visitors of Web pages/applications expect the page to load as quickly as possible, irrespective of the speed of their network or the capability of their device.

Along with quick loading, another important parameter is to make Web applications more responsive. If a page doesn't meet these two criteria, then users generally move out of it and look for better alternatives. So, both from the technical and economical perspectives, it becomes very important to optimise the responsiveness of Web pages.

Optimisation cannot be thought of just as an add-on after completing the design of the page. If certain optimisation practices are followed during each stage of Web page development, these will certainly result in a better performance. This article explores some of these best practices to optimise the performance of the Web page/application.

Web page optimisation is an active research domain in which there are contributions from so many research groups. An easy-to-use Web resource to start with the optimisation

of Web pages is provided by Yahoo (<https://developer.yahoo.com/performance/rules.html>). There are other informative resources, too, such as BrowserDiet (<https://browserdiet.com/en/#html>). Various other factors that contribute to Web page optimisation are shown in Figure 1.

Content optimisation

When responding to end user requests, the most time is taken up by the downloading of components such as images, scripts, Flash and style sheets.

- The greater the number of HTTP requests, the more the time required for the page to load and its responsiveness lessens. A critical mechanism to reduce the number of HTTP requests is to reduce the number of components in the Web page. This may be achieved by combining several components. For example, all scripts can be combined, many CSS style sheets can be merged together, etc.
- Minimising the DNS lookup is another important factor in optimisation. The primary role of Domain Name Systems is the mapping of human readable domain names to IP addresses. DNS lookups generally take somewhere between 20 and 120 milliseconds. Minimising the number of unique host names will reduce the number of DNS